

CONNOR F. CRUZ

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Education

Yale University — New Haven, CT
B.S. in Biomedical Engineering

Expected May 2030

North Carolina School of Science and Mathematics (NCSSM Online)

June 2026

- GPA 4.00/4.00. College-level coursework in Computational Science, Cryptography, Nanotechnology.

Charlotte Latin School — Charlotte, NC

June 2026

- GPA 4.29/4.33 unweighted (4.91 weighted); SAT 1600 superscore. National Merit Semifinalist; Cum Laude Society; Malone Scholar.

Technical Projects

E.M.I. (Engaged Memory Imaging) — Founder & Sole Developer

2025 – Present

JavaScript, React; originally JavaFX + Firebase

- Designed and built a face- and name-memorization app for people living with dementia, turning photographs of family members into low-friction recall exercises.
- Implemented three interaction modes: a concentration-style matching game, a browsable family gallery, and a multi-difficulty recall quiz. Rearchitected a stalled JavaFX prototype as a browser-based app.
- Ran an accessibility pass (large targets, high-contrast type, keyboard-navigable flows) for users with limited fine motor control.

Na'dañña' Guam — MIT Fab Academy Final Project

2024

Embedded C/Arduino, KiCad, Fusion 360; RP2040 + ATtiny1614

- Designed and fabricated a 16 × 24 in. interactive map teaching the culture of Guam: placing a 3D-printed latte-stone marker on a landmark drives a 4 in. TFT display through that site's history and photographs.
- Milled a custom PCB integrating a Seeed Xiao RP2040, an ATtiny1614, and an array of SS49E hall-effect sensors concealed beneath the board; wrote the sensing and display firmware.
- Solved a half-inch magnetic detection limit by inlaying the acrylic map into a CNC-milled pocket rather than surface-mounting it, making marker placement register reliably.

Research & Publications

NCSSM Journal of Student Computational Science

December 2025

- "Modeling Cultural Assimilation Using Global Migration, Economic Integration, and Cultural Diversity Metrics." Built and validated a computational model of assimilation dynamics from international migration data.

Technical Skills

Languages: Python, Java, C++ (advanced); JavaScript, HTML/CSS, LaTeX

Tools: React, Next.js, Firebase, Arduino, KiCad, Fusion 360, Git/GitHub, Vercel

Fabrication: CNC milling, laser & vinyl cutting, 3D printing, PCB design and production, embedded systems

Methods: Computational modeling, data analysis, cryptography, accessible design

Selected Programs & Awards

- **MIT Fab Academy**, Diploma Recipient (2024) — five-month intensive covering the full design-to-prototype pipeline; capstone was Na'dañña' Guam, above. ([project portfolio](#))
- **PROMYS**, Boston University (2024) and **MATHILY-ER**, Bryn Mawr College (2023) — residential summer programs (each admitting <10%) in number theory, proof writing, and discrete mathematics.
- **Scholastic Art & Writing Awards:** Gold Key, Personal Essay & Memoir (2025); Gold Key, Critical Essay (2024); Silver Key, Critical Essay (2025). **College Board National Recognition Program** (2024, 2025); AP Scholar with Distinction; Harvard Prize Book Award (2025).
- **National Latin & Greek Exams:** Gold Medal, Summa Cum Laude (Advanced Latin, Attic Greek, Intermediate Latin); CAMWS Latin Translation Exam Book Award, top 30 regionally (2025).

Leadership & Service

Math Club — Leader, Charlotte Latin School

2024 – 2025

- Led weekly problem-solving sessions and organized competition prep; member since 2022.

Mel's Diner & Meal Bags To Go — Volunteer, Charlotte, NC

2019 – 2026

- Prepared and distributed meals and pantry bags twice monthly for food-insecure residents of two North Carolina counties.